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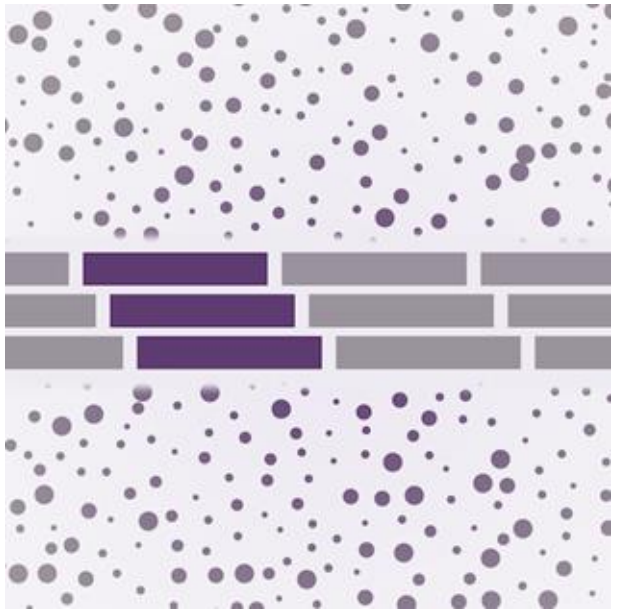
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Module 3: General Concepts in Genomics and Quality Control

Dr. Ido Hatam

CBW High-Throughput Genomics

November 4-6, 2026



Instructor affiliation logos go here

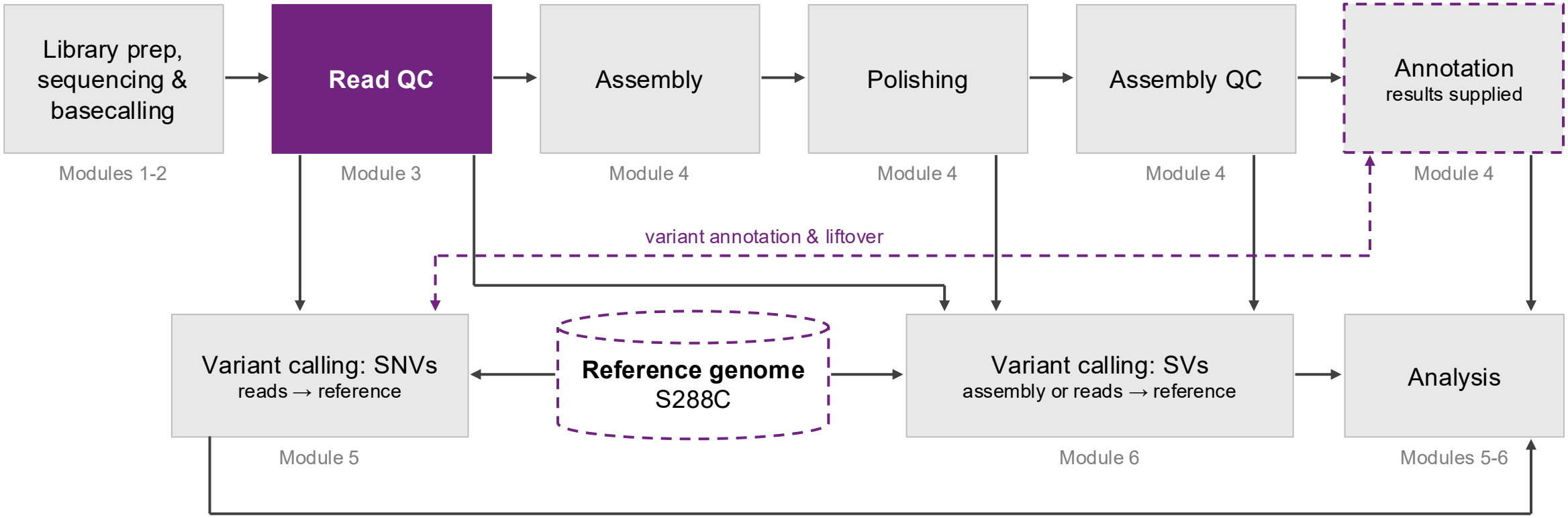
Learning Objectives

- Describe the main sources of error in **short reads** (Illumina) and **long reads** (ONT)
- Outline the common **genomics workflow** and place each workshop module on it
- Interpret **QC metrics** for short and long reads, and judge whether the data are fit for use

Goals for this workshop

- We do **comparative genomics**: we compare two yeast strains, one wild and one lab strain
- We start from raw sequencing reads, real and full size
- We build a genome from the reads, then find how it differs from the reference
- You run each step yourself, on a computing cluster
- **How do we compare genomes? The workflow**

Where we are in the workflow



= this module (dashed outline = supplied or reference)

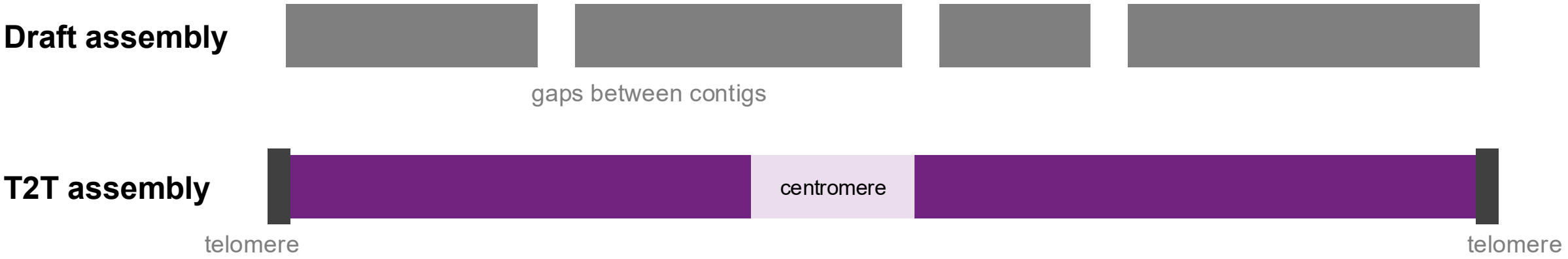
What is a genome and what are genomics

- A **genome** is the totality of an organism's hereditary information (DNA)
- It is more than the genes: it also includes the DNA between them
- **Genomics** is the scientific study of the genome: structure, function, evolution, mapping
- **Comparative genomics**: compare genomes to see what is shared and what differs
- Other “omes” exist (transcriptome, proteome); we start with the DNA layer

The standard reference genome: S288C

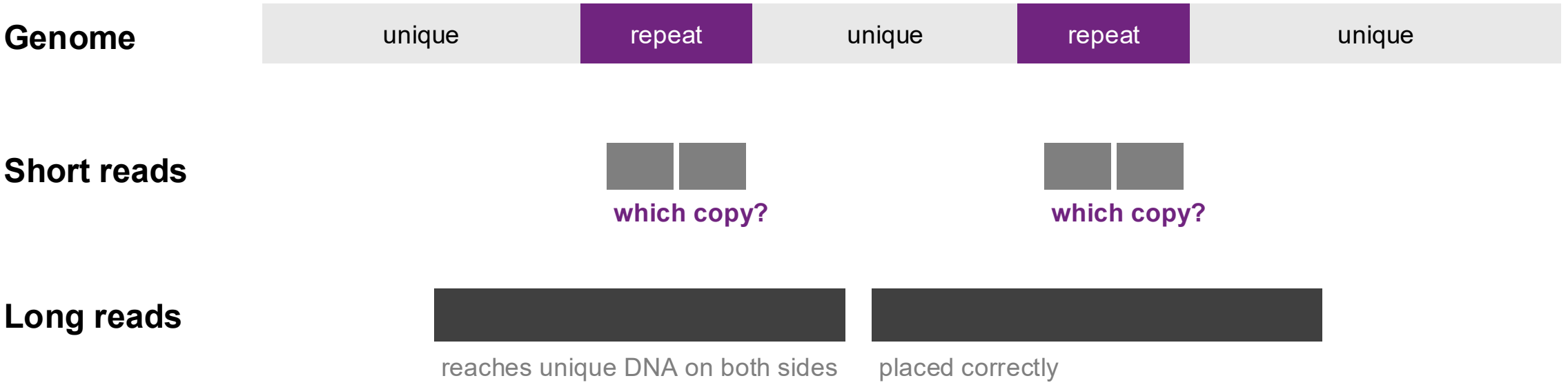
- A **reference genome** is an existing assembly used as a coordinate system
- For yeast the standard is S288C (release R64-1-1): the corgi, our lab-bred reference strain
- Designed by Mortimer as a lab strain; about 88% of its genome comes from EM93, a rotting fig isolate (California, 1938)
- Reads are mapped to it, variants are described on it, genes are named on it
- But one strain stands in for the whole species
- A wild isolate can differ in genes, rearrangements and repeats

What is a T2T assembly?



- **T2T**: each chromosome is one gapless sequence, telomere to telomere
- Includes centromeres and large repeats
- Long reads made it possible
- Human T2T-CHM13 completed the last ~8% of GRCh38 (Nurk et al. 2022)

Why long reads changed assembly

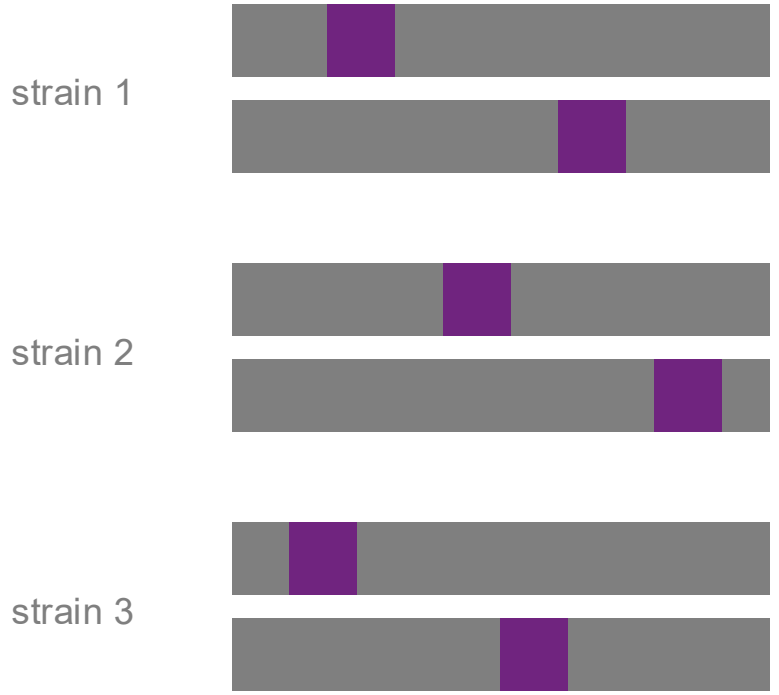


A short read inside a repeat fits every copy: it cannot be placed.

A long read that spans the repeat is placed with certainty.

From one genome to many

- One genome describes one individual
- **Phased** assemblies: both copies of each chromosome
- **Pangenome**: all sequence across many genomes (Liao et al. 2023)
- **ScRAP**: 142 T2T yeast genomes, several phased (diploids, polyploids)
- Captures the species' diversity, including SVs (O'Donnell et al. 2023)



two haplotypes per strain; purple = differences

S288C versus ScRAP T2T assemblies

	S288C reference	ScRAP T2T assemblies
Represents	One lab strain, for the whole species	Each strain's own genome
Chromosome ends and repeats	A few repeat regions collapsed (e.g. rDNA)	Chromosome ends assembled; several heterozygous strains phased
Strain-specific genes and SVs	Not visible	Captured
In this workshop	The coordinate system for variants	The goal for our own assembly

The workflow, stage by stage (1/2)

Stage	Goes in → comes out	Tools here	Module
1 Library prep, sequencing & basecalling	DNA → raw signal → FASTQ (bases + qualities)	Done for you	1-2
2 Read QC (quality control)	FASTQ → quality report; filtered FASTQ	sequali, filtlong, ShortRead	3
3 Assembly	Reads → contigs	Flye	4
4 Polishing	Contigs + reads → corrected contigs	medaka	4

The workflow, stage by stage (2/2)

Stage	Goes in → comes out	Tools here	Module
5 Assembly QC	Assembly → contiguity, completeness	QUAST, BUSCO	4
6 Annotation	Assembly → where genes are, what they do	funannotate, eggNOG-mapper (supplied)	4
7a Variant calling: SNVs	Reads + reference → SNVs	medaka, bcftools, snpEff	5
7b Variant calling: SVs	Assembly or reads + reference → SVs	minimap2 + SVIM-asm (lab); Sniffles (reads)	6
8 Analysis	Variants + genes → biology	IGV	5-6

The wolf: BAL_1a

- A **wild** *Saccharomyces cerevisiae* (baker's and brewer's yeast) isolate from rotten wood, Bawangling Mountain (Hainan, China)
- ENA: “a monosporic isolate from strain BAL/HN6”
- Monosporic: grown from a single spore; a homozygous diploid
- Sequenced with Oxford Nanopore long reads: run **ERR8562432**
- We use all of the reads, not a subsample, from QC to variants

Where our data come from



ScRAP: a project assembling *S. cerevisiae* strains from long reads

O'Donnell et al. 2023, *Nature Genetics*: T2T assemblies of 142 yeast strains

Anyone can look these IDs up on the **ENA** (European Nucleotide Archive)

The case: a wolf and a corgi

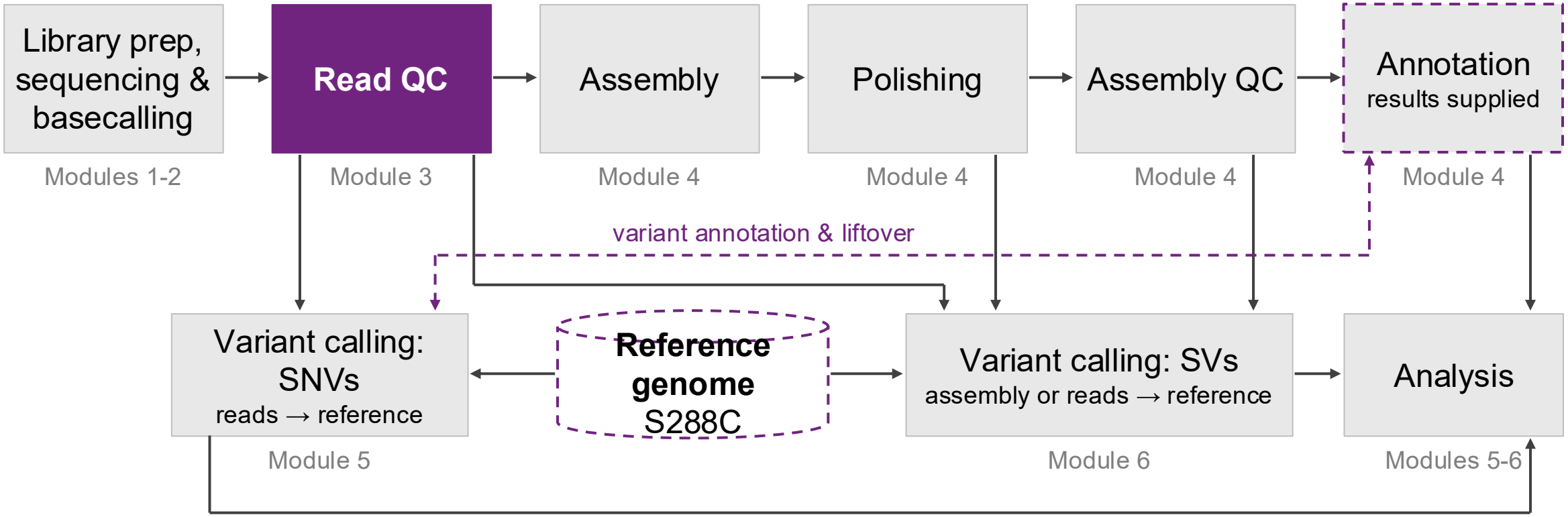
The wolf (our isolate, BAL_1a)
Wild; rotten wood, Hainan, China
Homozygous diploid, from one spore

The corgi (the reference strain, S288C)
Lab-bred; designed by Mortimer
About 88% from EM93, a 1938 fig isolate

We compare the two genomes: which variants separate them?
Two lines of questioning (thug life vs cushy desk job):

- Ethanol fermentation
- Stress response

Let's begin with our workflow



= this module (dashed outline = supplied or reference)

The case: are the wolf's reads (BAL_1a) good enough to use?

Our reads: ERR8562432

MinION ONT sequencer	**R9.4.1** pore type	**?** reads	**?** bases	**?** coverage

Coverage = total bases ÷ genome size. Illustrative example: 600 Mb ÷ 12 Mb = 50×

Library kit: SQK-LSK108 (older chemistry: expect lower quality)

The file name (BAL_1a.porechop.fq.gz) shows the reads were adapter-trimmed with Porechop

Adapter: a known synthetic sequence added in library prep; not genome

Reads, bases and coverage: you measure all three in Lab 3

The FASTQ format: four lines per read

Line 1	**@read_001**	Read ID (starts with @)
Line 2	G A T T A C A G G C **N** T T A A C C G G A T C G A	Bases: A, C, G, T or N
Line 3	**+**	Separator
Line 4	I I I I G G D D ? ? **#** ? ? : : 5 5 0 0 + + + & &	One quality character per base

Lines 2 and 4 always have the same length

N: a base the basecaller could not call

The Phred quality score (Q)

$$Q = -10 \times \log_{10}(P)$$ where P = the probability the base call is wrong

Q	Chance of error	Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%

Higher Q = more confident call

Every +10 means 10× fewer errors

Q starts at 0; the top value depends on the platform

Illumina aims for Q30+; nanopore: about Q8-Q13 (5-15% errors). Where do ours sit? Lab 3

Quality as one character: Phred+33

ASCII code = Q + 33 (so Q = code − 33)

! Q0 ASCII 33	**+** Q10 ASCII 43	**5** Q20 ASCII 53	**?** Q30 ASCII 63	**I** Q40 ASCII 73

One character per base, instead of two digits

Decode: “?” is ASCII 63, and 63 − 33 = 30, so **Q30**

Older files used +64; modern FASTQ uses +33

N: an ambiguous base

- **N**: the basecaller could not decide between A, C, G and T
- It writes N, usually with a very low quality score (Illumina: Q2 “#”)
- Many Ns make a read suspect
- Short-read pipelines often drop reads with any N
- Not the same as **N in an assembly**: runs of N mark gaps between joined contigs

N in a read



A, C, G or T? Unknown

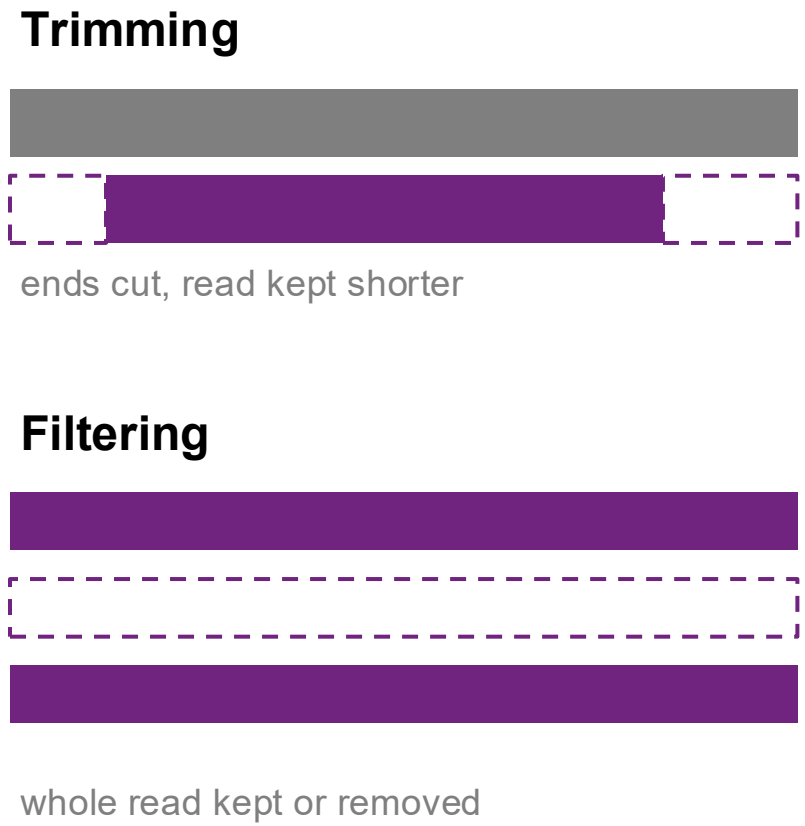
N in an assembly



gap of unknown sequence

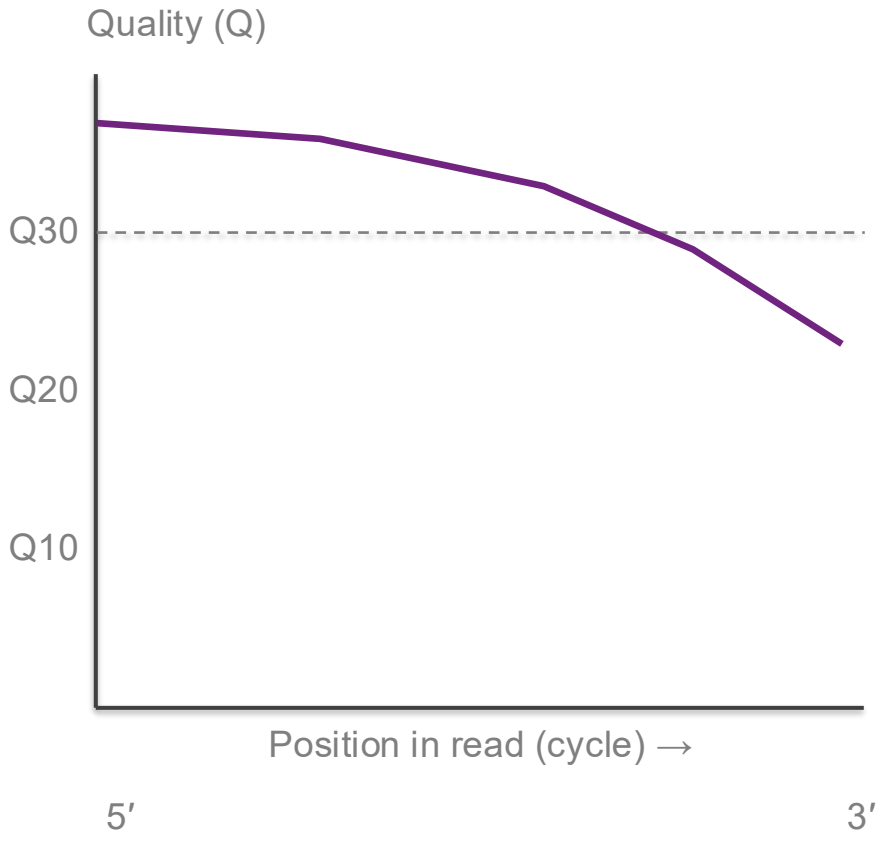
What is quality control (QC)?

- **QC**: measuring data quality before you use it
- Garbage in, garbage out
- **Filtering** removes whole reads
- **Trimming** cuts parts of reads
- Metrics: per-position quality, mean read quality, read length distribution, read N50, Ns
- Tools: ShortRead (Illumina), sequali and filtlong (ONT)



Recap: errors in short reads (Illumina)

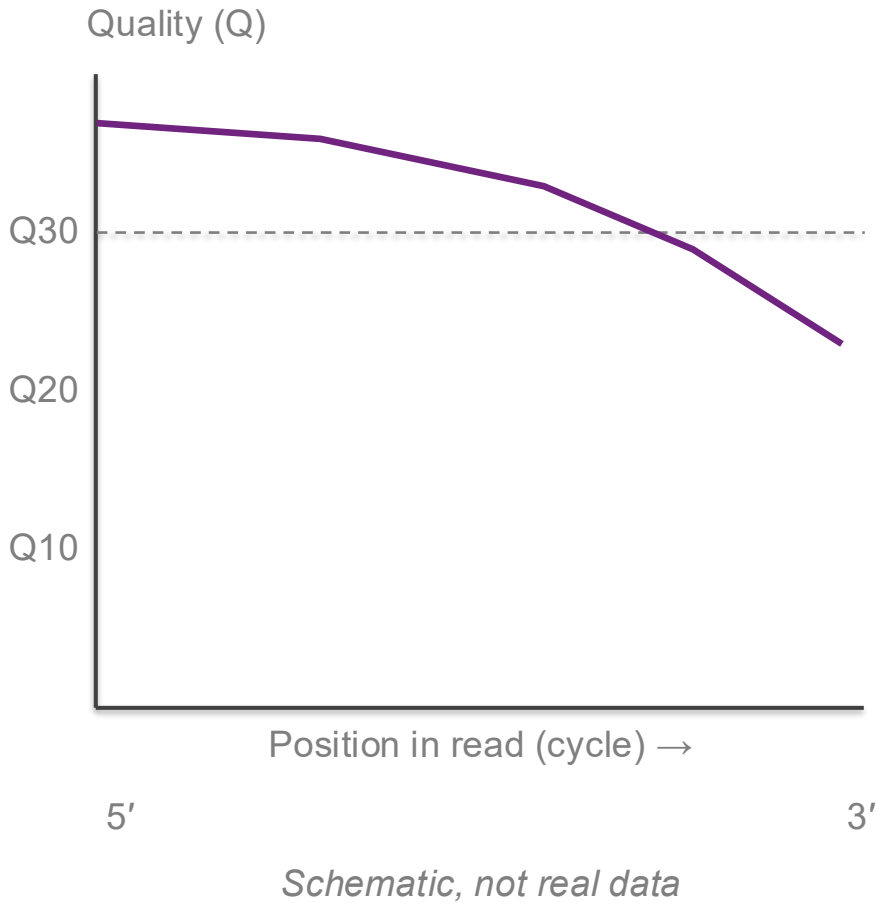
- Short reads: about 150-300 bp, accuracy above 99.9%
- Every copy in a cluster is read together, one base per cycle
- **Phasing**: some copies fall out of step (lead or lag), so the signal blurs
- Errors build up with each cycle: quality falls toward the **3′ end**
- Read 2 (R2) is usually worse than read 1 (R1)



Schematic, not real data

Recap: errors in short reads (Illumina)

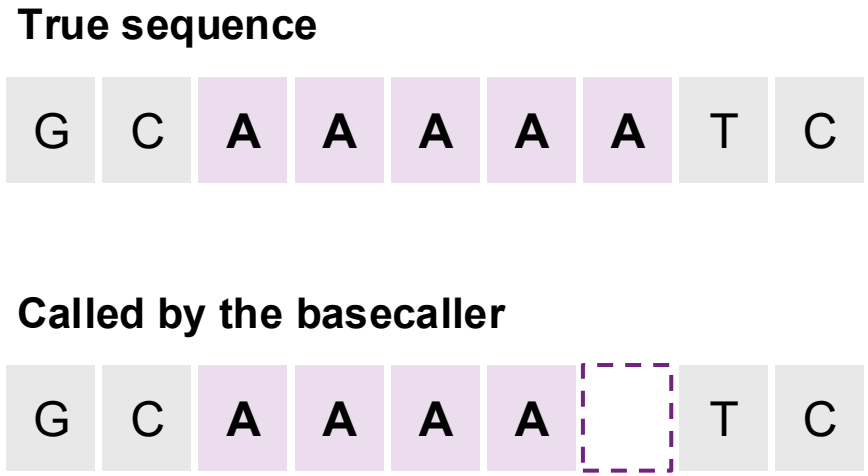
- Why do you think that R2 is worse?
- What does the error source tell you about the potential for Illumina to produce longer reads?



Schematic, not real data

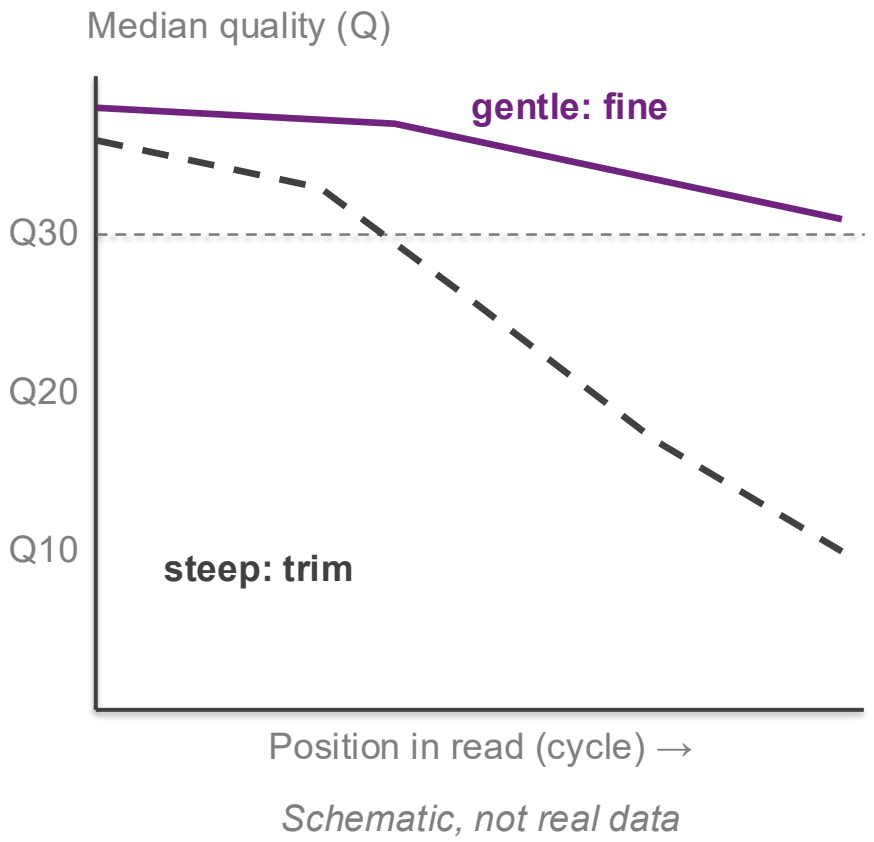
Recap: errors in long reads (ONT)

- Current changes as one DNA strand passes through a **nanopore**
- Several bases sit in the pore at once: the signal is a blurred mix
- **Homopolymers** (runs of one base): the count is hard to read
- Errors are mostly insertions and deletions (**indels**), and partly systematic
- About 5-15% errors. No decline along the read



Short reads: per-position quality

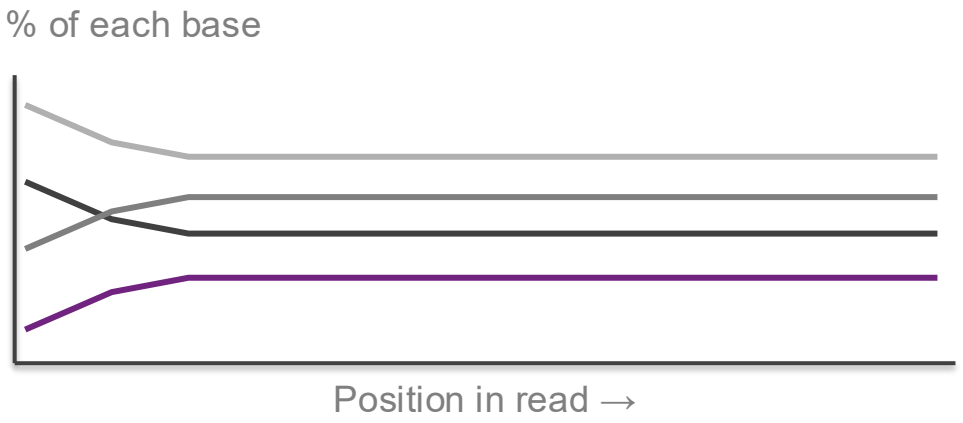
- **Per-position quality**: quality at each position along the read (a **cycle** is one position)
- Quality starts high and falls toward the 3′ end
- R2 is usually worse than R1
- Ask: where does the median cross **Q30**?
- A steep drop means trim the end, or a poor run



Schematic, not real data

Short reads: composition, duplicates

Base composition per cycle

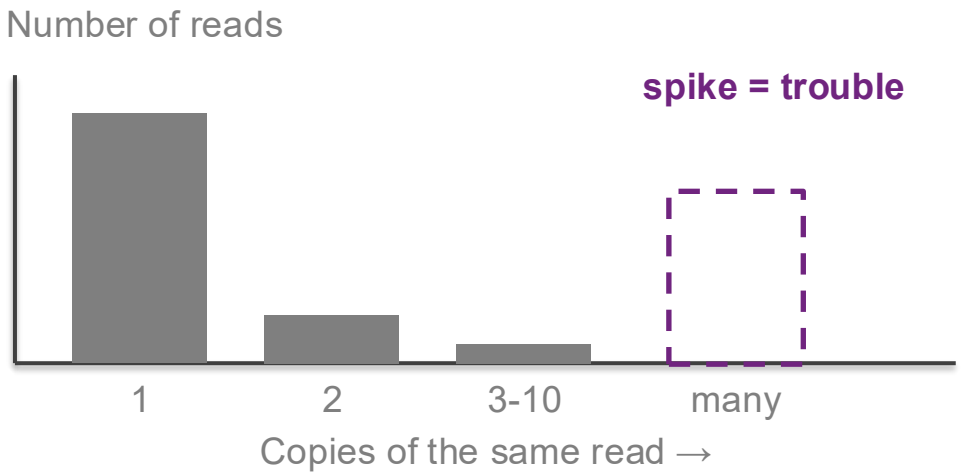


Expect flat, parallel lines; A ≈ T and G ≈ C, at levels set by the genome's GC (yeast: A, T ≈ 31%; G, C ≈ 19%)

Wobble in the first ~10 bases is common

Spikes or split lines: adapters or GC bias

Read duplication



Most reads should appear once

A spike of identical reads: possible PCR bias

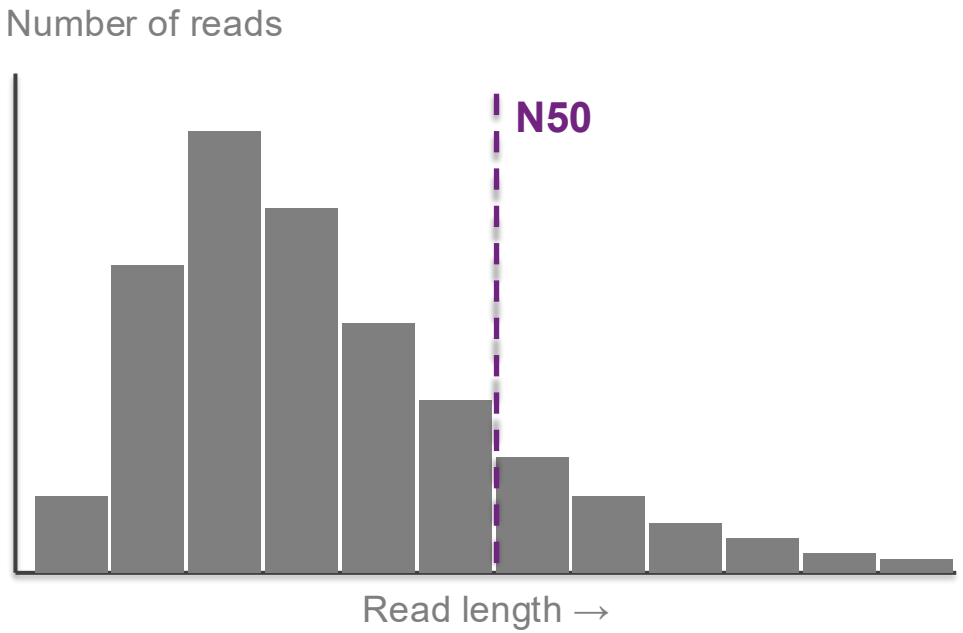
Also check: reads with **N**

Long reads versus short reads

	Short reads (Illumina)	Long reads (ONT)
Read length	Fixed length, 150-300 bp	A distribution: hundreds of bp to over 100 kb
Quality along the read	Falls toward the 3′ end	Roughly stable, no 3′ drop
Typical median quality	Q30 or higher	About Q8-Q13 (5-15% errors); about Q20 (newest ONT chemistry)
Files per sample	Two (R1 and R2)	One FASTQ

Read lengths and read N50

Read length distribution (schematic)

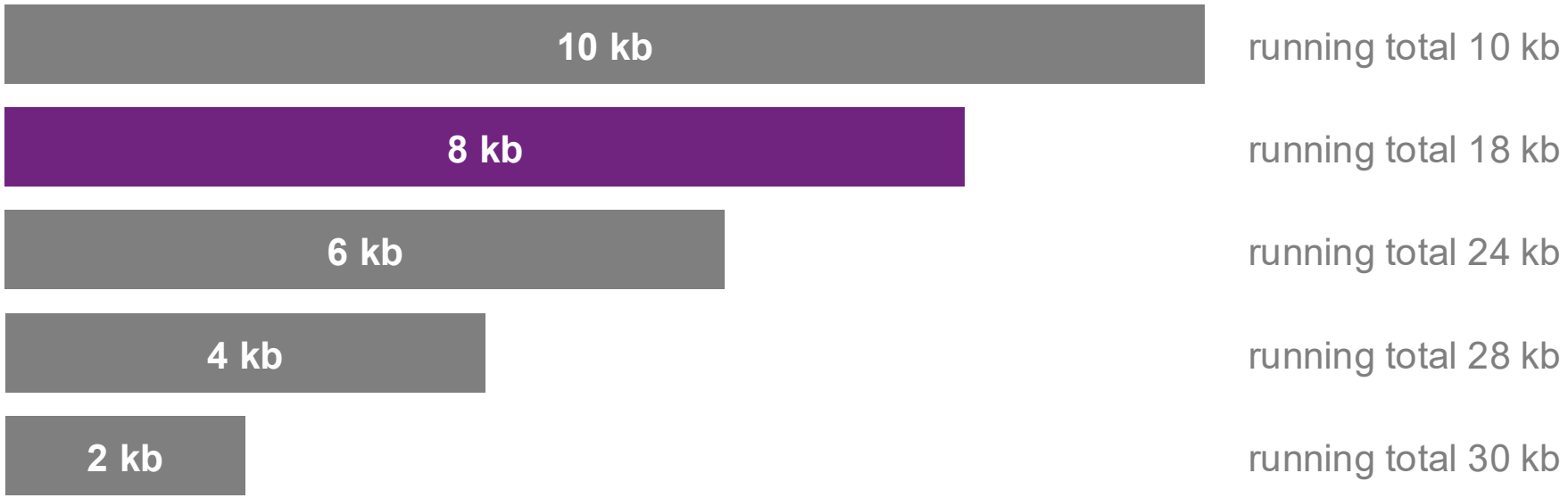


Many shorter reads, a tail of very long ones

Read N50
A length-weighted median: 50% of all bases are in reads at least this long

Example: read N50 = 2 kb
means half of your bases sit in reads of 2 kb or longer

Read N50: a worked example



All reads: 30 kb. Half: **15 kb**. 10 kb is not enough; 10 + 8 = 18 kb passes it

Read N50 = 8 kb (the mean is 6 kb)

Mean read quality

A read with 10 bases



Average the Q values: Q20 (looks good)

Average the error probabilities:
(0.001 + 0.1) ÷ 2 = 0.0505
→ about **Q13**

The poor bases dominate, so use this version

Your turn in Lab 3
Which average does sequali's per-read quality use? Look at its histogram in Lab 3

Read QC metrics: what and why

Metric	What it shows	Why it is useful
Read length distribution	How many reads at each length	Is the library mostly long reads or short fragments?
Per-position quality	Quality at each position along the read	Finds stretches of poor quality that trimming might target
Base composition	% of A, C, G, T at each position	Flat lines expected; bumps point to adapters or bias
GC content	% of G + C per read	One peak near 38% (S288C); extra peaks suggest contamination
Ns	Share of N bases	Many Ns make a read suspect

sequali, our long-read tool, reports all of these

Are the data fit for use?

Ask	Short reads	Long reads (ONT)
Overall quality as expected?	Median above Q30	About Q8-Q13 (5-15% errors)
Quality steady along the read?	Small 3′ drop; steep drop: trim	Should be stable
Lengths as expected?	One fixed length	Broad spread; check read N50
Few Ns?	Ideally none	Few; many make a read suspect
Enough data?	Coverage is enough	Coverage is enough (aim for 30-50×)

Fit for use = as expected for the platform, stable, and enough data

Trim or filter?

	Short reads	Long reads
Where errors are	Build up at the 3′ end	Scattered along the read; vary from read to read
Trimming	Trim adapters and 3′ ends below ~Q20; then filter short reads and reads with N	Nuanced: no one-size-fits-all recipe
Our approach	Trim, then filter	**Filter, don't trim** (a rule of thumb)

Trimming shortens long reads, and length is their strength

Tools you may meet with short reads: QC and trimming

Tool	What it does
FastQC	QC report per FASTQ file
fastp	QC plus adapter and quality trimming in one step
Cutadapt	Adapter (and quality) trimming

Not covered in this workshop (our bonus uses ShortRead)

Filtering trade-off: filtlong

```
filtlong --min_length 1000 --keep_percent 90 reads.fastq.gz | gzip > filtered.fastq.gz
```

Setting	What it does
--min_length 1000	Drops reads shorter than 1,000 bp
--keep_percent 90	Keeps the best 90% of bases, ranking reads by length and quality

Gain: less junk in the assembly

Cost: fewer bases, lower coverage

In the lab: **not run**. Work out what it would remove, then assemble the raw reads.

Summary

- Short reads lose quality toward the 3′ end
- ONT errors are mostly indels, often at homopolymers
- QC is stage 2 of the workflow
- FASTQ: four lines per read; Q = −10 $\log_{10}$(P error)
- Check lengths, read N50, per-position and mean read quality, GC, Ns
- Judge data against expectations; long reads: filter, don't trim

Lab 3 preview: what you will do

1 Set up folders (nano, .sh)	**2** Download reads and reference	**3** Check the tools	**4** Run sequali with sbatch
5 Copy the report with scp	**6** Think through filtlong	**7** Submit the Flye assembly, then lunch	**Bonus** ShortRead on Illumina data

Questions are about reading results, not writing code

Coffee Break & Networking Session

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